

Quiz 1: Agents and Search

Weeks 1-2

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Weeks 1-2

Points

20 total

Expected time

30 minutes

Submission

Individual PDF

Instructor verification checklist

- ▶ Work independently.
- ▶ Show intermediate frontier or arithmetic steps.
- ▶ State every assumption used for a guarantee.
- ▶ No electronic communication.

Solution 1.a - Agent formulation (2 points)

Question focus. Give one element for each PEAS component.

Answer. Performance: timely collision-free deliveries; environment: aisles, shelves, people, and stations; actuators: wheel and lift commands; sensors: lidar, odometry, and task messages.

Solution 1.b - Agent formulation (2 points)

Question focus. Explain why rational does not mean omniscient.

Answer. A rational agent maximizes expected performance from its percept history and knowledge. It can still fail because observations and outcomes are uncertain; omniscience would require knowing actual future consequences.

Solution 2.a - BFS and DFS trace (2 points)

Question focus. Trace BFS until a goal is removed.

Answer. FIFO trace: [S], then [A,B], then [B,G], then [G,C], then remove G. The returned path is S-A-G.

Solution 2.b - BFS and DFS trace (2 points)

Question focus. Trace DFS when successors are pushed in alphabetical order.

Answer. After pushing A then B, LIFO removes B, then C, then G; it can return S-B-C-G. A different documented push convention earns credit if traced consistently.

Solution 3.a - Cost-sensitive search (2 points)

Question focus. Which route can BFS return, and why is that not a minimum-cost guarantee?

Answer. BFS returns the depth-two route S-A-G with cost 6, because it minimizes depth under FIFO order rather than nonuniform path cost.

Solution 3.b - Cost-sensitive search (2 points)

Question focus. Trace UCS priorities until the goal is removed.

Answer. UCS removes S(0), B(1), C(2), G(3), and returns S-B-C-G with cost 3.

Solution 3.c - Cost-sensitive search (2 points)

Question focus. State conditions under which UCS is complete and optimal.

Answer. With finite branching and step costs bounded below by a positive epsilon, UCS is complete and removes an optimal goal first.

Solution 4.a - A-star heuristic (2 points)

Question focus. Define f , g , and h in A-star.

Answer. g is accumulated path cost, h estimates optimal remaining cost, and $f=g+h$ orders the frontier.

Solution 4.b - A-star heuristic (2 points)

Question focus. Prove Manhattan distance is consistent.

Answer. A legal step costs one and changes Manhattan distance by at most one, hence $h(n)$ is at most $1 + h(n\text{-prime})$ for every edge.

Solution 4.c - A-star heuristic (2 points)

Question focus. If two nodes have $(g,h)=(4,3)$ and $(5,1)$, which is removed first?

Answer. Their f values are 7 and 6, so the second node is removed first, subject only to stale-entry handling and ties.

Edge-case reasoning and expected behavior

- ▶ Return the empty path.
- ▶ Use graph-search bookkeeping to terminate.
- ▶ Apply the stated deterministic secondary key.

Grading rubric

Category	Points	Full-credit evidence
Agent formulation	4	PEAS and rationality are precise.
BFS and DFS trace	4	FIFO/LIFO order and trace are correct.
Cost-sensitive search	6	Depth/cost distinction and UCS update are correct.
A-star heuristic	6	Admissibility, consistency, and f values are correct.

Reflection exemplar

Answer. A suitable answer identifies finite branching, uniform costs for BFS optimality, or a positive lower bound on step cost for UCS, with the relevant algorithm named.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.